

Assignment 1:
Machine Learning - Analytical Reasoning.

Group 5:

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1. Introduction

Machine learning (ML) is a field of artificial intelligence, and a core part of modern data analytics. ML enables computational systems to understand & learn meaningful patterns from historical data and make predictions without being explicitly programmed for every decision. In healthcare sector, these methods are especially valuable because they assist in identifying patients who may be at higher clinical risk at an early stage. This enables healthcare providers to make better decisions regarding prevention and treatment.

This “analytical reasoning” assignment focuses on binary classification using the *Clinical_Risk_v2* dataset. Based on risk-status, each patient is classified into only two categories: high-risk or low-risk. The dataset includes a range of demographic, behavioral, and physiological attributes such as age, blood pressure, cholesterol level, BMI, smoking status, and family history.

The main objective is to provide clear analytical reasoning and justification for identifying which modelling approach is both statistically accurate and clinically appropriate, rather than focusing only on implementation. Only two interpretable models, Logistic Regression and Decision Trees, are used based on the assignment requirement. Principal Component Analysis (PCA) is incorporated only where its application can be theoretically justified. Furthermore, the model effectiveness is evaluated using multiple metrics: accuracy, precision, recall, and F1-score to ensure balanced assessment as misclassification in healthcare can lead to unequal and potentially serious outcomes in clinical settings.

Overall, this report applies conceptual understanding and critical analysis to compare machine learning models for clinical risk prediction and evaluate their practical trade-offs.

2. Part 1 - Conceptual Understanding

2.1 Is this problem likely to be linearly separable?

The *Clinical_Risk_v2* dataset is unlikely to be linearly separable because clinical risk does not depend on a single clean factor. Instead, it is influenced by a combination of interacting factors such as age, blood pressure, cholesterol level, BMI, and lifestyle behaviors. These factors rarely combine in a strictly linear manner. For example, moderately high blood pressure may be a low-risk factor for a younger patient but

high-risk for an older patient. This indicates non-linear interactions between features. The dataset also includes categorical variables (e.g., smoking status, physical activity, registration month) that introduce non-ordinal patterns which cannot be captured by a single linear boundary. These characteristics suggest that a simple linear model cannot fully capture the complexity of feature interactions in the dataset. As a result, non-linear models may be more appropriate for this problem.

2.2 Two informative features and one uncertain/misleading feature

Informative Feature 1: Age

Age is clinically a meaningful predictor of risk. In the given *Clinical_Risk_v2* dataset, most high-risk patients are older. This aligns with established medical evidence that cardiovascular and metabolic risks increase with age. Its continuous nature and strong theoretical relevance make it a reliable feature.

Informative Feature 2: Blood Pressure

Blood pressure is directly linked to cardiovascular risk and shows higher values among many high-risk patients in the dataset. As a physiological measure with clear clinical interpretation, it contributes meaningfully to distinguishing risk categories.

Uncertain/Misleading Feature: Registration Month

The registration month has no physiological connection to clinical risk. Any observed pattern may be connected to administrative activities or seasonal factors. Inclusion of this feature may introduce noise or spurious correlations and it will potentially mislead the model training.

2.3 Why accuracy alone is insufficient

Relying only on accuracy can be misleading and can cause serious health issues. Because, it does not capture the different consequences of prediction errors in a clinical setting. For instance, in a false negative situation, a high-risk patient is wrongly classified as low risk. This can delay the treatment of a patient which will lead to serious health outcomes. In contrast, a false positive is comparatively less harmful. This would only trigger additional screening. Also, accuracy treats both types of errors with equal importance. As a result, this does not reflect real-world clinical urgencies. Additionally, for example, if the dataset is imbalanced, a model may appear highly accurate simply by predicting the

majority class. On the other hand, the model may fail to detect many high-risk patients. For this reason, metrics such as precision, recall, and F1-score provide a more reliable and clinically relevant evaluation in this context.

3. Part 2 - Controlled Modeling

3.1 Train/Test Split

We used a fixed 80/20 train-test split to ensure consistent evaluation across all models. The split was stratified on the target variable “risk_status” to preserve the proportion of high-risk and low-risk patients in both sets. To ensure reproducibility, a random_state value of 42 was applied.

```
# Splitting the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Figure 1: Train-test split

3.2 Models Training

- **Logistic Regression**

A Logistic Regression classifier was trained using a preprocessing pipeline that applied:

- StandardScaler to numeric features and
- OneHotEncoder to categorical features

This ensured all features were appropriately normalized and encoded before model fitting.

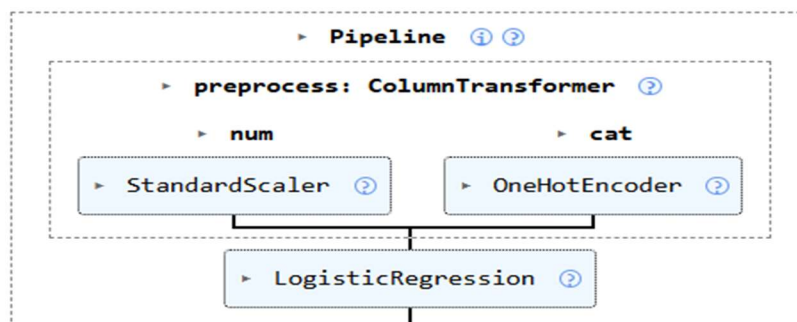


Figure 2. Logistic Regression model pipeline

- **Decision Tree Classifier**

A Decision Tree model was trained using the same preprocessing pipeline. Decision Trees do not require feature scaling, but using the same pipeline ensures consistency and also prevents data leakage.

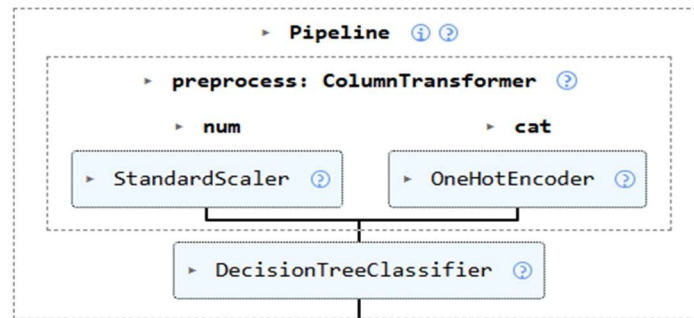


Figure 3. Decision Tree model pipeline

- **Application of Principal Component Analysis (PCA)**

PCA with 2 components was applied after preprocessing. The purpose is to reduce the high-dimensional one-hot encoded feature space into a compact representation.

Justification for PCA:

- One-hot encoding expands categorical variables into many binary columns.
- PCA compresses these correlated dimensions into a smaller set of orthogonal components.
- This helps explore whether dimensionality reduction improves model performance or stability.
- PCA also enables visualization and comparison of model behavior in a reduced feature space.

Two additional models were trained on the PCA-transformed data:

- Logistic Regression (PCA)
- Decision Tree (PCA)

```

# Apply PCA (2 components) on the preprocessed training data

# First fit the preprocessor on training data
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)

# Apply PCA
pca = PCA(n_components=2)
X_train_pca = pca.fit_transform(X_train_processed)
X_test_pca = pca.transform(X_test_processed)
  
```

Figure 4. PCA transformation (2 components)

```
# Train Logistic Regression on PCA-transformed data
log_reg_pca = LogisticRegression(max_iter=1000)
log_reg_pca.fit(X_train_pca, y_train)

y_pred_log_pca = log_reg_pca.predict(X_test_pca)

log_pca_accuracy = accuracy_score(y_test, y_pred_log_pca)
log_pca_precision = precision_score(y_test, y_pred_log_pca)
log_pca_recall = recall_score(y_test, y_pred_log_pca)
log_pca_f1 = f1_score(y_test, y_pred_log_pca)

print(f"log_pca_accuracy: {log_pca_accuracy}")
print(f"log_pca_precision: {log_pca_precision}")
print(f"log_pca_recall: {log_pca_recall}")
print(f"log_pca_f1: {log_pca_f1}")
```

Figure 5. Logistic Regression with PCA results

```
# Train Decision Tree on PCA-transformed data
dt_pca = DecisionTreeClassifier(random_state=42)
dt_pca.fit(X_train_pca, y_train)

y_pred_dt_pca = dt_pca.predict(X_test_pca)

dt_pca_accuracy = accuracy_score(y_test, y_pred_dt_pca)
dt_pca_precision = precision_score(y_test, y_pred_dt_pca)
dt_pca_recall = recall_score(y_test, y_pred_dt_pca)
dt_pca_f1 = f1_score(y_test, y_pred_dt_pca)

print(f"dt_pca_accuracy: {dt_pca_accuracy}")
print(f"dt_pca_precision: {dt_pca_precision}")
print(f"dt_pca_recall: {dt_pca_recall}")
print(f"dt_pca_f1: {dt_pca_f1}")
```

Figure 6. Decision Tree with PCA results

3.3 Evaluation Metrics

All models were evaluated on the test set using:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a balanced view of performance. The combination carries utmost importance especially in a clinical context where false negatives and false positives carry different risks.

Table 1: Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.89	0.9125	0.948	0.9299
Decision Tree	0.87	0.9211	0.9091	0.915
Logistic Regression (PCA)	0.77	0.77	1.0	0.87
Decision Tree (PCA)	0.62	0.7671	0.7273	0.7467

Logistic Regression achieved the highest overall performance. Decision Tree performed slightly lower but remained competitive. PCA reduced overall performance stability, especially for the Decision Tree, due to loss of feature-level detail. Logistic Regression was more stable under PCA, while the Decision Tree was more sensitive to dimensionality reduction.

4. Part 3 - Comparative Reasoning

4.1 Sensitivity to feature representation:

Between the two models, the Decision Tree is generally more sensitive to the way features are represented since it splits data using thresholds on individual variables. Even small changes in encoding or scaling can influence the overall structure of the tree. One-hot encoding creates many sparse binary columns, and PCA transforms features into new combinations. These disrupt the tree's natural splitting behavior. Logistic Regression is more stable because it operates in a continuous weight space and handles scaled or one-hot encoded features smoothly. Its performance depends on the overall geometry of the data, not on specific feature thresholds.

4.2 How PCA Affects Each Model: Mechanism:

Logistic Regression: PCA helps to reduce multicollinearity by creating orthogonal components. This aligns well with linear models. However, reducing the data to only two components removed important clinical information which caused a slight decline in performance.

Decision Tree: PCA harms the Decision Tree much more. Trees rely on clear, interpretable thresholds (e.g.: blood_pressure > 130). PCA rotates the feature space into abstract combinations, removing these thresholds and breaking the structure the tree depends on. This explains the sharp decline in performance.

4.3 Failure Modes of Each Model

Logistic Regression: Fails when relationships are non-linear. If the true pattern involves interactions (e.g.: high BMI plus high cholesterol together increase risk), Logistic Regression is not capable of capturing this. It needs manual addition of interactions.

Decision Tree: Fails due to overfitting. Trees memorize training data by creating deep, highly specific splits. This leads to high variance, poor generalization and instability with small changes in data.

4.4 Clinical Interpretation

Which error is more costly?

In a clinical risk-prediction setting:

- False Negative (FN) = predicting “low risk” when the patient is actually high risk
- This is the most dangerous error because it may delay treatment or monitoring.

Which model handles this better?

Logistic Regression handles false negatives better because:

- It achieved the highest recall (0.948)
- High recall means it correctly identifies more high-risk patients
- It is more stable and less sensitive to noise
- It generalizes better than the Decision Tree

Therefore, for clinical safety, Logistic Regression is the safer model because it minimizes false negatives more effectively.

5. Part 4 - PCA as a Decision

5.1 Was PCA Necessary?

PCA was not essential for this dataset. Even with one-hot encoding, the models performed well in the original feature space. Importantly, there was no major multicollinearity problem. However, applying PCA was still useful in this analysis. It allowed us to observe how the models behave when the information is compressed. It also helped assess whether their performance remains stable under reduced dimensions. In other words, PCA added analytical value even if it wasn't required for accuracy.

5.2 What Did PCA Improve or Harm?

What has PCA improved:

- It removed correlation among features by creating orthogonal components.
- Logistic Regression benefited slightly because it prefers independent predictors.
- It reduced noise from many sparse one-hot encoded columns.

What has PCA harmed:

- Reducing everything to two components caused major information loss.
- Decision Tree performance dropped sharply because PCA removes the clear, interpretable thresholds that trees rely on.
- Even Logistic Regression lost some accuracy, showing that compression came at a cost.

Overall, PCA made the data cleaner but less meaningful, and the trade-off hurt performance, especially for the Decision Tree.

5.3 Alternative Approach to PCA

A more suitable approach for this dataset is feature selection rather than feature compression. Methods such as SelectKBest, mutual information, or L1-regularization reduce dimensionality while keeping the original clinical variables intact. This avoids the information loss caused by PCA and preserves the clear, interpretable thresholds that models especially Decision Trees depend on. Feature selection keeps the structure of the data meaningful, reduces noise, and supports both models more effectively than transforming everything into abstract components.

6. Conclusion

This analysis showed that Logistic Regression is the most reliable model for predicting high-risk patients. The model offers the strongest recall and F1-score under a consistent 80/20 split and unified preprocessing pipeline. Decision Tree performed reasonably well, but was far more sensitive to how the features were represented. Once the PCA was applied, its accuracy dropped sharply. Because, it lost the clear and interpretable clinical thresholds that it normally relies on to make splitting decision. PCA was useful for understanding how the models react to compressed information. However, it did not improve predictive performance. For this dataset, feature selection or regularization would be a more suitable dimensionality-reduction strategy. Overall, Logistic Regression without PCA provides the best balance of accuracy, stability, and clinical safety.

7. References

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8. AI Tool Use Disclosure

Artificial intelligence tool (Copilot) was used only for supporting the generation and formatting of academic references in this report. No AI tools were used for data analysis, model development, interpretation of results, or writing of the main content. All analytical reasoning, explanations, and conclusions were developed independently.